


Photonic Integrated Circuits

# Silicon Microring Resonator Study

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# 1 Introduction

Silicon photonics has become one of the most promising photonic integration platforms in the last years. This can be attributed to its high index contrast and to the capacity to leverage commercial complementary metal-oxide semiconductor (CMOS) fabrication infrastructure to implement integrated photonic circuits (PICs) [1]. This permits fabrication companies to focus on the process developments while design companies focus on the technical design and applications [2].

One interesting technology developed in this context and applicable to wavelength division multiplexing (WDM) are micro-ring resonators. WDM is necessary to scale up the capacity in optical communications and interconnects, it is critical to implement robust low-cost silicon photonic WDM transmitters and receivers [2].

Therefore, this project aims to study how to dimension a micro-ring resonator for WDM applications based on performance requirements and fabrication process limitations.

# 2 Physical Description

In principle, a ring resonator consists of a looped optical waveguide and a coupling mechanism to access the loop. When the waves in the loop build up a round trip phase shift that equals an integer times $2\pi$, the waves interfere constructively and the cavity is in resonance [3].

Thus, to describe the resonance condition of this cavity, we must characterize how light propagates in this dielectric optic waveguide. First, to confine the light inside the waveguide with total internal reflection, the refractive index of the core must be higher than that of the cladding. Si is commonly used in the core, since it has $n = 3.47$ at $\lambda = 1550$ nm, and silica in the cladding, with $n = 1.45$. Additionally, well chosen cross-section dimensions are essential to permit single-mode operation.

The vectorial wave equations that describe the field in the waveguide will be solved here through the effective index method (EIM), which gives a good working approximation. It consists in solving successive approximations of a two-dimensional (2D) rectangular waveguide by one-dimensional (1D) slab waveguides, which can be separately analyzed.

This gives as a set of wave equations for TE and TM polarizations which can be solved to determine the effective index $n_{eff}$.

Due to dispersion effects, the parameter used to quantify the total dependence of the effective index on wavelength is the group index, which is defined as:

$$n_g = n_{eff} - \lambda_0 \frac{\partial n_{eff}}{\partial \lambda} \tag{1}$$

For an accurate determination of the modes and bending loss in a 3D bent waveguide, however, we must resort to a full-vectorial analysis, requiring a numerical solution of the vector wave equation, which is typically formulated in a local cylindrical coordinate system [4]. Many commercial software can use FDTD or eigenmode solvers to calculate the group index from the given geometry of the ring and the desired operating wavelength.

In this study, we will be adopting the add-drop ring resonator configuration, shown in figure 1.

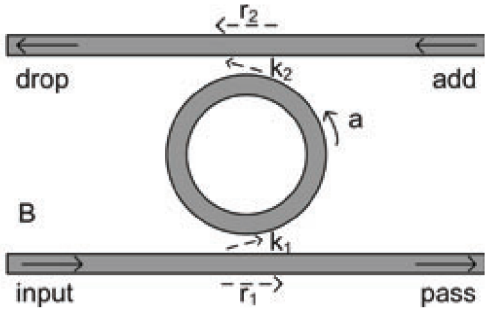



Figure 1: Add-drop ring resonator configuration. Source: [3]

An important factor to determine now is the coupling between the ring and the straight waveguides that have the input and output signals. Figure 2 presents a model of the edge-to-edge separations between the waveguides. The separation $s$ is then given by equation 2.

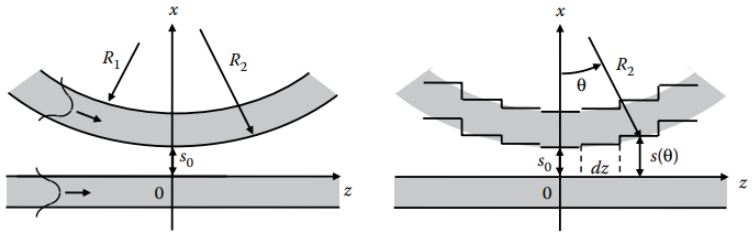



Figure 2: Bent waveguide model for coupling calculations. Source: [4].

$$s(\theta) = s_0 + R_2(1 - \cos\theta) \tag{2}$$

And, $k_c(s)$ being the coupling strength between two straight waveguides separated by a gap $s$, the total field coupling between the waveguides will be:

$$k = \int_{-\pi/2}^{\pi/2} k_c(s) R_2 \cos\theta \, \mathrm{d}\theta \tag{3}$$

Given these cross-coupling field amplitudes, the power spectral responses at the drop port and through port can be obtained as ($\phi_{rt} = 2\pi\beta R$ is the roundtrip phase and $a_{rt} = e^{-\alpha\pi R}$ is the roundtrip field attenuation factor) [4]:

$$T_d = \frac{a_{rt} k_1^2 k_2^2}{1 + a_{rt} r_1^2 r_2^2 - 2 a_{rt} r_1 r_2 \cos\phi} \tag{4}$$

$$T_t = \frac{r_1^2 + a_{rt}^2 r_2^2 - 2 a_{rt} r_1 r_2 \cos\phi_{rt}}{1 + a_{rt} r_1^2 r_2^2 - 2 a_{rt} r_1 r_2 \cos\phi_{rt}} \tag{5}$$

Observe that $r_1$ and $r_2$ are the field self-coupling coefficients.

If the attenuation is negligible ($a \approx 1$), critical coupling occurs at symmetric coupling ($k_1 = k_2$). For a lossy resonator, critical coupling occurs when the losses match the coupling as $r_2 a = r_1$ [3]. This condition ensures zero through-port transmission at resonance.

Observing these equations, we can see that wavelengths for which the roundtrip phase is a multiple of $2\pi$ will have a response peak in the drop port and a valley in the through port [5].

The distance between two of these resonances is called free spectral range (FSR) and is given by:

$$FSR = \frac{\lambda^2}{Ln_g} \tag{6}$$

$L$ is the length or the ring resonator, encompassing both the bent sections and possible straight sections in what is called a 'racetrack' configuration. Therefore, a lower ring radius is interesting in the sense that it leads to a higher FSR. However, we must consider that a smaller radius leads to higher losses in the waveguide [3].

Another important expression we can derive for the add-drop ring resonator is the full width at half maximum (FWHM), related to the sharpness of the filter. It is given by:

$$FWHM = \frac{(1 - r_1 r_2 a)\lambda_{res}^2}{\pi n_g L \sqrt{r_1 r_2 a}} \tag{7}$$

Related relevant metrics are finesse and Q-factor:

$$Finesse = \frac{FSR}{FWHM} \tag{8}$$

$$Q = \frac{\lambda_{res}}{FWHM} \tag{9}$$

# 3 Design

To comply with the single-mode condition at $\lambda = 1.55\ \mu m$, the cross section dimensions of the ring waveguide are generally on the sub-micrometer range. The height is commonly in the 200 to 300 nm range and the width varies between 400 and 500 nm for SOI waveguides [3].

As a known example, we will consider a waveguide with 220 nm height and 450 nm width with an Si core and a $SiO_2$ cladding, which has a group index of $n_g \approx 4.3$ for $\lambda = 1.55\ \mu m$ [3]. To confirm this assumption, we model the waveguide in GDSFactory and use tidy3D's mode solver to calculate the electric field of the TE mode 0 inside the waveguide, shown in figure 3. Additionally, the calculated group index was of 4.297, extremelly close to the predicted value.

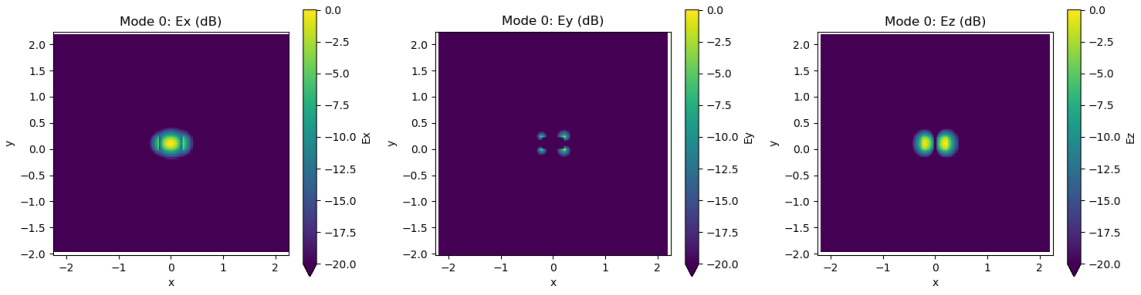



Figure 3: TE mode electric field inside the waveguide.

We can then choose a 5 $\mu m$ bending radius, which will have a low bending loss (of 2.7 dB/cm according to the simulation code). Then, we can define a total length of about 65 $\mu m$ for the total ring, which will give a theoretical $FSR = 8.7$ nm (or 1.1 THz). We could, for

example, fit five 200 GHz channels in this range. Simulating the bent waveguide with the setup previously described, we get a group index of $n_g = 4.289$ and an effective index of $n_{eff} = 2.382$.

Then, the total length can be adjusted in a few tens of nanometers to choose the desired resonant wavelength ($\lambda_{res} = Ln_{eff}/m$). We choose $m = 99$, which leads us to choose a length of 16.48 $\mu m$ for the straight sections of the racetrack ring. Modeling the final structure in GDSFactory, we get figure 4.

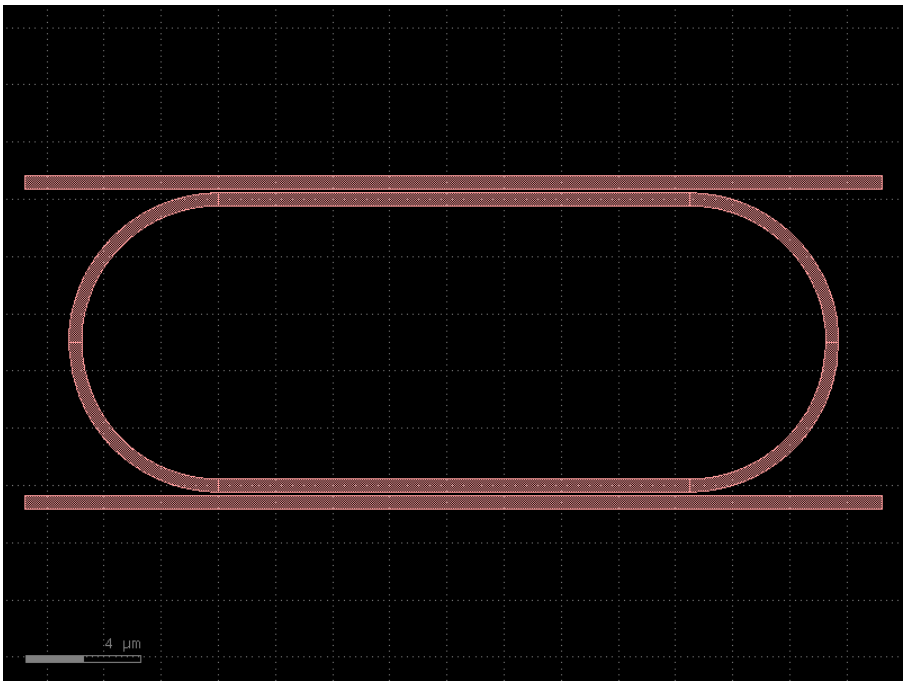



Figure 4: Add-drop ring structure modelled in GDSFactory.

Note that the resonance wavelength of the ring is very sensitive to deviations in the fabrication process. A 1 nm deviation in width can lead to a 0.8 nm deviation in the resonance wavelength. Thus, an integrated heater deposited near the ring waveguide to permit active tuning of the resonant wavelength through the thermo-optic effect and thermal expansion. It allows large wavelength tunability, which can compensate the resonance variations from both fabrication and environmental temperature fluctuations [2].

We should choose the gap between the waveguides in the coupling region seeking to maximize the on-resonance drop transmission subject to the required passband width and out-of-band rejection. In this case, a gap of 150 nm was chosen. The long straight sections also contribute to increase coupling.

Figure 5 shows the theoretical power spectral response on the drop and through ports of the microring resonator modeled in this section (according to equations 4 and 5). We may observe that the FSR agrees with the expected value and that the drop port has a reasonably good frequency selectivity. However, the passband lacks uniformity and may distort large bandwidth signals.

To get more precise results, we now simulate the GDSFactory model shown in figure 4 and simulate it using a 2.5D FDTD simulation in Meep. The 2.5D consists in building a 2D version of the 3D structure by calculating a vertical effective index for its materials and then performing a 2D FDTD with it. It is less precise, but greatly increases performance.

In our case, we calculated the effective index for each wavelength of the simulation to

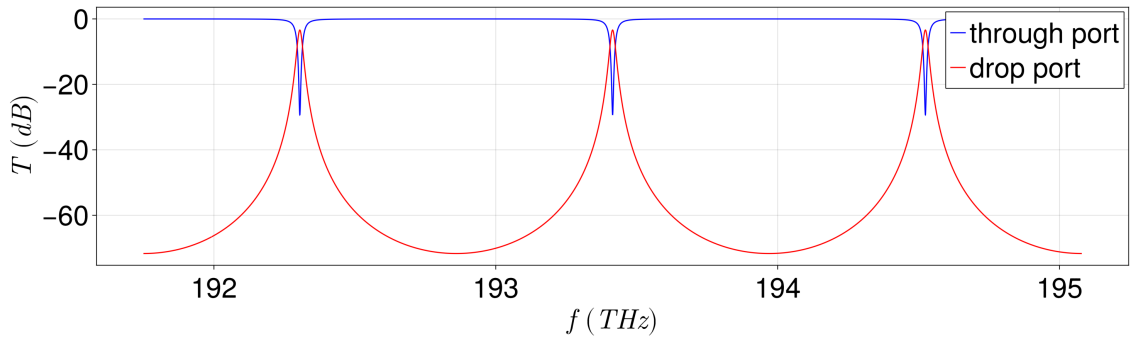



Figure 5: Power spectral response of the modeled ring resonator (in theory).

account for dispersion. The resulting transmission characteristics of the resonator calculated by the simulation are shown in figure 6. Ideally, more wavelength points should be taken to see the exact resonance peak characteristics, but we can still observe that the resonant wavelength is close to the desired 1550 nm, and that the FSR also approaches the value predicted by theory.

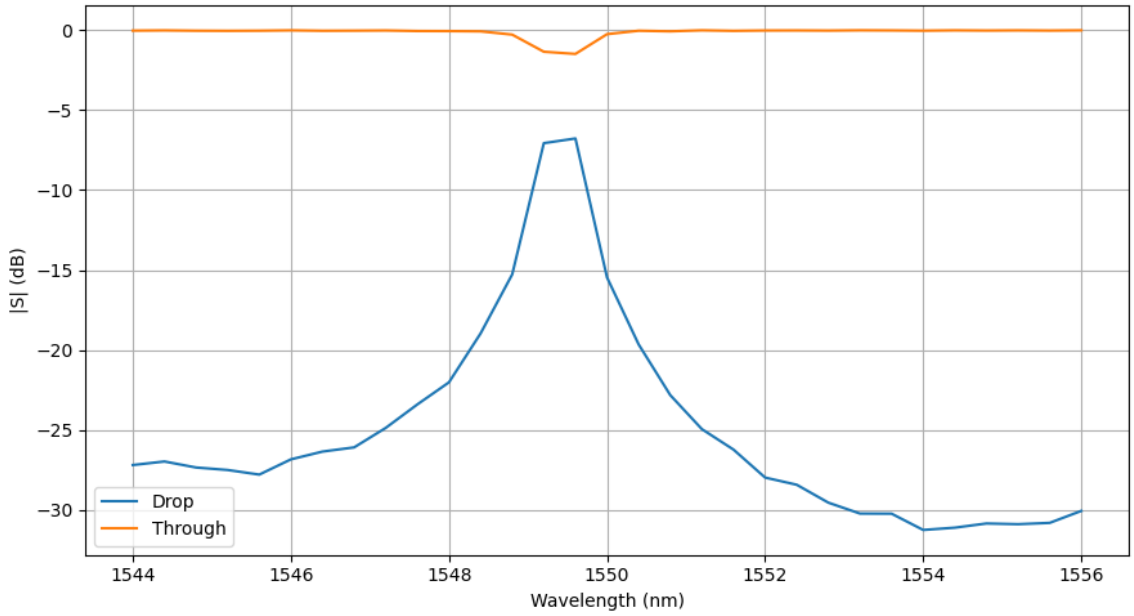



Figure 6: Power spectral response of the simulated ring resonator.

Through a sequence of ring resonators where each ring acts as a filter for a different wavelength, a mux/demux for WDM can be built.

# 4 Conclusion

Although highly sensible to fabrication defects and environmental changes, microring resonators can play a useful role as spectral filters for communication purposes to multiplex or demultiplex WDM signals.

# References

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